



Geographic and host-mediated population genetic structure in a cestode parasite of the three-spined stickleback

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Comparative studies of genetic diversity and population structure can shed light on the ecological and evolutionary factors that influence host–parasite interactions. Here we examined whether geography, time and genetic variation in Alaskan three-spined stickleback (*Gasterosteus aculeatus* Linnaeus) hosts shape the population genetic structure of the diphyllbothridean cestode parasite *Schistocephalus solidus* (Müller, 1776). Host lineages and haplotypes were identified by sequencing the mitochondrial cytochrome *b* gene, and host population structure was assessed by Bayesian clustering analysis of allelic variation at 11 microsatellite loci. Parasite population structure was characterized according to allelic variation at eight microsatellite loci. Mantel tests and canonical redundancy analysis were conducted to evaluate the proportion of parasite genetic variation attributable to time and geography vs. host lineage, haplotype, and genotypic cluster. Host and parasite population structure were largely discordant across the study area, probably reflecting differences in gene flow, environmental influences external to the host, and genomic admixture among host lineages. We found that geography explained the greatest proportion of parasite genetic variation, but that variation also reflects time, host lineage, and host haplotype. Associations with host haplotypes suggest that one parasite genotypic cluster exhibits a narrower host range, predominantly infecting the most common host haplotypes, whereas the other parasite cluster infects all haplotypes equally, including rare haplotypes. Although experimental infection trials might prove otherwise, distributional differences in hosts preferentially infected by *S. solidus* could underlie the observed pattern of population structure. © 2016 The Linnean Society of London, *Biological Journal of the Linnean Society*, 2016, 00, 000–000.

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INTRODUCTION

Population genetic structure and genetic diversity can be governed by stochastic factors such as those attributable to geography and deterministic factors that give rise to selective pressures (Endler, 1986; Slatkin, 1987; Avise, 2000). For parasites, population structure may be determined by host selection and host behaviour, in which parasite demes are hierarchically nested within individual hosts and host populations (Price, 1980; Gandon *et al.*, 1998; Rohani &

Ruxton, 1999). Thus, conditions arising from the host environment as well as conditions external to the host can potentially structure genetic variation in parasites (Dybdahl & Lively, 1996; Orsini *et al.*, 2013; Franke *et al.*, 2014).

Comparative approaches can shed light on whether genetic variation in parasites is shaped by host characteristics (Dybdahl & Lively, 1996; Jarne & Théron, 2001; Light & Hafner, 2008). For example, comparisons of topological and temporal patterns in host and parasite phylogenies can serve as tests for coevolution (Huelsenbeck, Rannala & Yang, 1997; Hafner & Nadler, 1990; Light & Hafner, 2008). Testing for

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concordant population genetic structure in a single host and parasite can indicate whether variation has been shaped by a shared geography (Dybdahl & Lively, 1996; McCoy, Bouludier & Tirard, 2005; Criscione, 2008), as has been observed in the parasite *Plagioporus shawi* and salmonid hosts across Oregon and Washington (Criscione & Blouin, 2007). Conversely, patterns of discordance may reflect asymmetries in gene flow, effective population sizes, and life history attributes (Mulvey *et al.*, 1991; McCoy *et al.*, 2005; Keeney *et al.*, 2009). For instance, broadly congruent population genetic structure occurs in the field mouse *Apodemus sylvaticus* and its nematode parasite *Heligmosomoides polygyrus*; but the parasite exhibits greater population genetic differentiation, possibly due to lower rates of gene flow or barriers not shared by the field mouse host (Nieberding *et al.*, 2004).

There is compelling evidence that host characteristics can define genetic variation in parasites with direct life cycles (Price, 1980; Hafner *et al.*, 1983; Blouin *et al.*, 1995; Nadler, 1995; Criscione & Blouin, 2004), but the importance of host characteristics is less clear for parasites with complex life cycles (Criscione & Blouin, 2007; Criscione, 2008; Štefka, Hypša & Scholz, 2009). Host attributes may have negligible or no influence on parasite genetic variation if, for example, movement potential strongly differs between intermediate and definitive hosts. Signatures of adaptive specificity to intermediate hosts can be obscured by gene flow driven by the vagility of definitive hosts (Jarne & Théron, 2001; Keeney *et al.*, 2009). Recent work on the stickleback–cestode model system (Heins & Baker, 2008; Barber, 2013) suggests, however, that the vagility of definitive hosts may have less influence on gene flow than expected based solely on movement potential (Nishimura *et al.*, 2011; Sprehn *et al.*, 2015). Experimental infection and concordant phylogenetic structure among stickleback hosts and *Schistocephalus* cestode parasites indicate that *S. solidus* is a specific parasite of three-spined sticklebacks (Bråten, 1966; Chubb *et al.*, 1995; Nishimura *et al.*, 2011; Franke *et al.*, 2014; Kalbe *et al.*, 2016). Evidence of intraspecific differentiation also suggests that genetic variation in *S. solidus* might track evolutionary divergence within three-spined sticklebacks (Kalbe *et al.*, 2016). Atlantic basin three-spined stickleback and Pacific basin stickleback are infected by evolutionary lineages of *S. solidus* that exhibit ~ 5% mitochondrial sequence divergence (Nishimura *et al.*, 2011). Similarly, comparisons of *S. solidus* collected from hosts in Oregon and Alaska recovered signatures of divergence (i.e., 1% sequence divergence; Nishimura *et al.*, 2011). Several genotypic clusters and significant genetic structure also have been

found across south-central Alaska, possibly due to specificity to the intermediate host or factors limiting definitive host dispersal (Sprehn *et al.*, 2015).

Variation in three-spined stickleback host traits could give rise to sub-specific host selectivity in *S. solidus* (Kalbe *et al.*, 2016). Restricted gene flow combined with local selection and drift (Bell, 1976; Ortí *et al.*, 1994; Mäkinen, Cano & Merilä, 2006; Marques *et al.*, 2016) has given rise to a remarkable diversity in morphology and life history in three-spined stickleback (Hagen & McPhail, 1970; Bell, Aguirre & Buck, 2004; Kristjánsson, 2005). The two major stickleback mitochondrial (mtDNA) lineages found across the northern Pacific Rim (Haglund, Buth & Lawson, 1992; Ortí *et al.*, 1994; Johnson, Taylor & Harbour, 2004), for example, exhibit phenotypic (e.g., armoring and spinal reduction) variation that could promote intraspecific host specificity in *S. solidus* parasites (Deagle, Reimchen & Levin, 1996). Reproductive (Heins, 2012; Heins & Baker, 2014) and physiological differentiation (e.g., hormone signalling and MHC II variation; Kitano *et al.*, 2011; Eizaguirre *et al.*, 2011) could also give rise to adaptive specificity (though see Franke *et al.*, 2014).

Thus far, no tests have been undertaken to determine whether genetic variation in *S. solidus* is attributable to factors external (i.e., geographic isolation, temporal variation) or internal (i.e., genetic variation) to three-spined stickleback hosts. The goal of this study was to interrogate *a posteriori* inferences derived by Sprehn *et al.* (2015) to determine whether population genetic structure in *S. solidus* across south-central Alaska reflects common dispersal constraints or adaptive specificity to stickleback hosts. To do so, we derived new data on the genetic variation of Alaskan populations of freshwater three-spined stickleback hosts for comparison to available data on corresponding *S. solidus* parasites. This enabled us to determine whether stickleback hosts and *S. solidus* parasites exhibit concordant genetic structure (Nishimura *et al.*, 2011). It also enabled us to perform association analyses to disentangle the influence of geography and time from the influence of host genetic variation on *S. solidus* population genetic structure.

MATERIAL AND METHODS

STUDY SYSTEM

Three-spined stickleback host lineages

Repeated colonization from marine and anadromous sources, as well as depopulation and recolonization from Pleistocene glacial cycling, has given rise to several evolutionary lineages of freshwater three-spined

sticklebacks (Bell, 1976; Ortí *et al.*, 1994; Mäkinen *et al.*, 2006). The two evolutionary lineages that occur across Alaska (and elsewhere in the northern Pacific Rim) exhibit approximately 2.5% mtDNA divergence (Haglund *et al.*, 1992; Ortí *et al.*, 1994; Johnson *et al.*, 2004; Lescak *et al.*, 2015), which is comparable to levels of divergence among closely related species of other North American fishes (e.g., Blum *et al.*, 2008). The most prevalent lineage in Alaska, referred to as the Euro-North American Clade (ENAC), encompasses the Eastern Pacific basin and extends across Western North America as well as Europe (Ortí *et al.*, 1994; Johnson *et al.*, 2004; Lescak *et al.*, 2015). The other lineage is the Trans-North Pacific Clade (TNPC, also referred to as Japanese stickleback), which occurs only in the Pacific basin in and near the Sea of Japan and Okhotsk Sea, as well as in some Alaskan lakes (Deagle *et al.*, 1996; Thompson, Taylor & McPhail, 1997; Johnson *et al.*, 2004; Lescak *et al.*, 2015).

Parasite life cycle

Schistocephalus solidus exhibits a complex life cycle with cyclopoid copepods as the first intermediate host, three-spined stickleback as the second intermediate host, and one of over 40 species of piscivorous birds as the definitive host (Smyth, 1962). Parasite eggs are excreted from the definitive host, giving rise to coracidia, the only free-living stage of *S. solidus* (Smyth, 1962; Barber & Scharsack, 2010). Coracidia are consumed by several species of copepods, where they develop into proceroids (Smyth, 1962; Hammerschmidt & Kurtz, 2007; Barber & Scharsack, 2010). Copepods bearing proceroids are then consumed by three-spined sticklebacks, after which proceroids penetrate the gut wall to invade the body cavity of the host fish, transform into plerocercoids, and increase dramatically in mass using host nutrient reserves (Smyth, 1962; Hammerschmidt & Kurtz, 2007; Barber & Scharsack, 2010). Shortly after fish are ingested by the definitive host, the plerocercoid attains sexual maturity and begins reproducing. Eggs are dispersed into lakes or streams with defecation (Smyth, 1962).

SAMPLE COLLECTIONS AND DNA EXTRACTIONS

Infected three-spined sticklebacks were collected between 2009 and 2012 from 15 lakes in three regions of south-central and south-west Alaska: the Matanuska-Susitna Valley (MatSu), the Kenai Peninsula (Kenai), and the Bristol Bay (BB) drainage (Fig. 1). Ten lakes were sampled in MatSu, three lakes in Kenai, and two lakes in BB (Table 1; Fig. 1). Collections were specifically approved by annual Fish

Resource Permits from the Alaska Department of Fish and Game and animal care protocols from Tulane University Institutional Animal Care and Use Committee (protocols 0304R-UT-C and 0304R2) and the University of Washington Institutional Animal Care and Use Committee (protocol 3142-01). Fish were captured using 3 mm and 6 mm wire-mesh minnow traps, euthanized with tricaine methanesulfonate (MS-222; Ostrander, 2000), and preserved in 95% ethanol. Fish were then dissected and *S. solidus* parasites were removed from each host and preserved separately in 95% ethanol. DNA extractions from three-spined sticklebacks and corresponding parasites were conducted using the DNeasy Blood and Tissue Extraction Kit (Qiagen, Inc., Valencia, CA, USA). Individual parasites were genotyped by Sprehn *et al.* (2015). Fish containing one or more parasites that were successfully genotyped were sequenced and genotyped for the current study.

MTDNA SEQUENCING AND ANALYSIS FOR STICKLEBACK HOSTS

We amplified a fragment of the cytochrome *b* (*cytb*) mitochondrial gene for 161 individuals using primers F 5'-ATGAACTTTGGTTCCCTCC-3' and R 5'-CGCTGAGCTACTTTTGCATGT-3' (Mäkinen & Merilä, 2008). PCR reactions were carried out in 15 µL reaction volumes with 10× *HoTaq* buffer, 0.3 µM primers, 0.5 mM MgCl₂, 0.33 mM dNTPs, and 2.5 units *HoTaq* DNA polymerase (MCLAB, Inc.). We ran PCRs on Eppendorf Mastercycler pro S thermocyclers (Eppendorf, Hauppauge, NY, USA) with the following reaction conditions: 10 min hot start activation at 94 °C followed by 30 cycles of 94 °C for 30 s, 52 °C for 45 s, and 72 °C for 60 s, with a final step of 72 °C for 10 min. Amplified PCR products were purified with ExoSAP-IT (Affymetrix, Cleveland, OH) before conducting cycle sequencing reactions using the Big Dye Terminator v. 3.1 kit (Applied Biosystems, Foster City, CA, USA) and sequencing primers F 5'-TTTGCTTTTCACTTCT-TATT-3' and R 5'-GCTCTAATAACCCCTTGGC-3' (Mäkinen & Merilä, 2008). Following purification with Sephadex (Sigma-Aldrich, St. Louis, MO, USA), sequencing reactions were run on an ABI 3730xl DNA Analyzer (Applied Biosystems, Foster City, CA, USA). Resulting chromatograms were assembled and aligned using Sequencher v. 4.9 (Gene Codes Corp, Ann Arbor, MI, USA).

After haplotypes were characterized and enumerated, haplotype diversity was calculated using Genalex v. 6.501 (Peakall & Smouse, 2006, 2012). Nucleotide diversity was calculated with Arlequin v. 3.5 (Excoffier & Lischer, 2010). We also used Arlequin v. 3.5 to estimate pairwise Φ_{ST} values of genetic

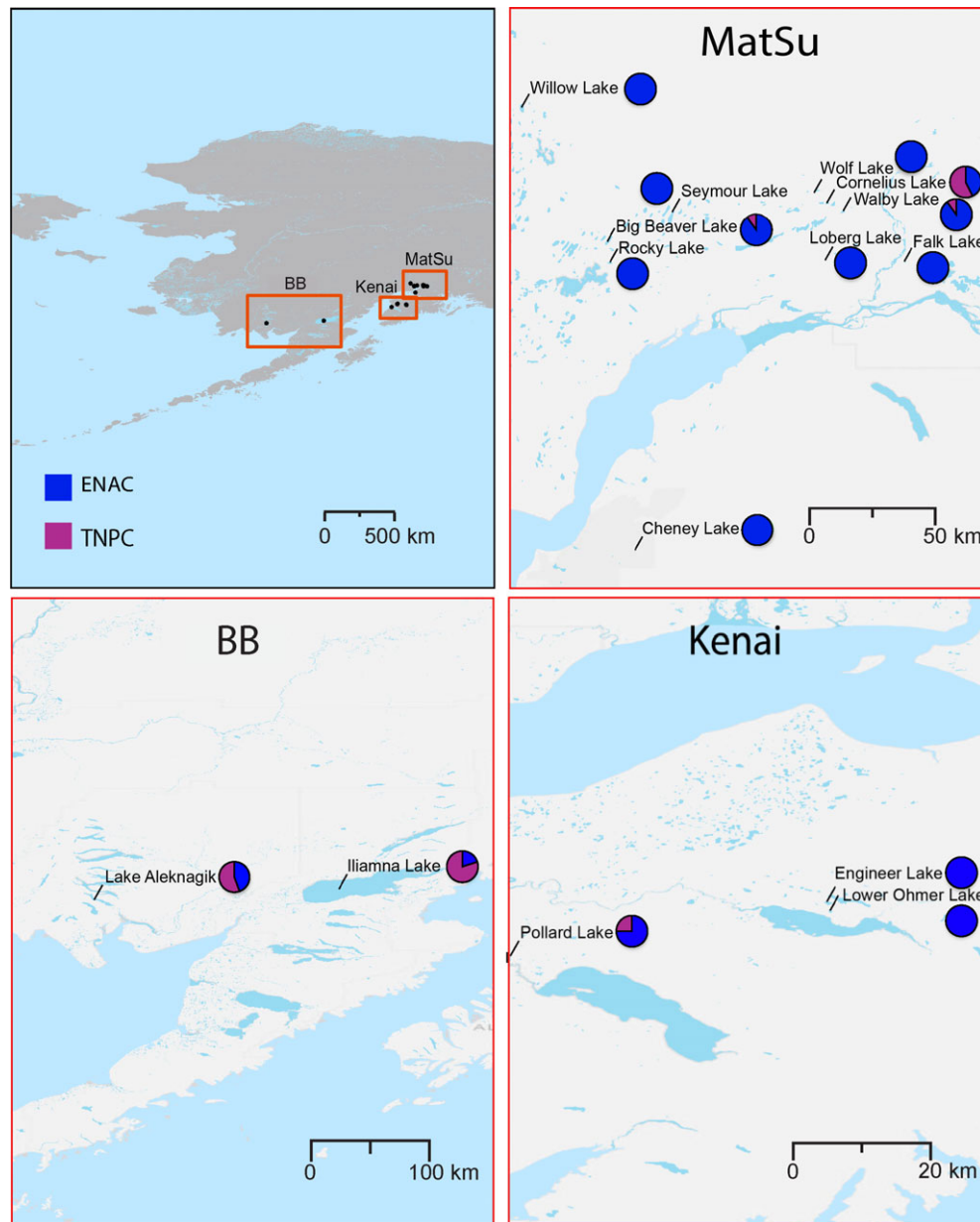


Figure 1. Distribution of three-spined stickleback lineages across the study area. Overview of the study region spanning the Matanuska-Sustina Valley (MatSu), Bristol Bay (BB), and the Kenai Peninsula (Kenai), with corresponding insets of each region (top right, bottom left, bottom right). Pie charts indicate the proportion of individuals belonging to the ENAC and TNPC lineages.

differentiation between lake populations and to conduct AMOVAs to test for hierarchical genetic differentiation across regions, lakes, and individuals, and among sampling years. The significance of pairwise population Φ_{ST} values was assessed by 10 000 bootstrap replicates and sequential Bonferroni correction for multiple comparisons (Holm, 1979). Genealogical relationships among haplotypes were inferred from a Median-joining haplotype network (Bendelt, Forster

& Röhl, 1999) constructed in Network v. 4.61 (www.fluxus-engineering.com). We also used MEGA v. 6.06 (Tamura *et al.*, 2013) to build a neighbour-joining tree based on Tamura-Nei distances, which was identified as the best substitution model in MEGA v. 6.06. Node support was assessed from 1000 bootstrap replicates. One *Pungitius pungitius* sequence (GenBank accession no. AF356080) was included as outgroup.

Table 1. Sampling and genetic diversity of three-spined stickleback hosts and *Schistocephalus solidus* parasites

Sampling	Host mtDNA variation					Host μ sat variation					Parasite μ sat variation						
	Region	Lat (N)	Long (W)	Years	N hosts	Total hap	H _d	N _d	H _o	H _e	N _a	A _r	parasites	H _o	H _e	N _a	A _r
Aleknagik	BB	59.214	158.532	2012	18	5	0.296	0.02	0.616	0.644*	6.364	2.51	39	0.626	0.649	7.875	3.65
Iliamna	BB	59.305	155.233	2012	10	3	0.213	0.015	0.681	0.695	6.364	2.78	39	0.614	0.732	9.375	4.16
Pollard	Kenai	60.194	151.141	2012	4	2	0.223	0.019	0.614	0.58	3.545	2.63	4	0.563	0.578	3.500	3.5
Engineer	Kenai	60.285	150.193	2012	10	2	0.006	0	0.615	0.642*	4.636	2.63	81	0.664	0.753	11.625	4.31
Lower Ohmer	Kenai	60.272	150.186	2012	22	2	0.003	0	0.471	0.536*	5.636	2.24	26	0.604	0.710	8.125	4.09
Willow	MatSu	61.444	150.033	2011, 2012	10	3	0.019	0.001	0.696	0.71	5.818	2.81	67	0.620	0.713	9.625	4.04
Big Beaver	MatSu	61.345	149.496	2011, 2012	10	4	0.126	0.009	0.614	0.701*	5.545	2.76	69	0.635	0.742	10.000	4.14
Rocky	MatSu	61.332	149.493	2011, 2012	10	3	0.03	0.002	0.775	0.677	5.182	2.69	45	0.658	0.735	7.500	3.98
Cheney	MatSu	61.121	149.454	2010, 2012	7	4	0.021	0.002	0.708	0.740*	7.273	2.98	82	0.666	0.730	10.250	4.09
Seymour	MatSu	61.37	149.394	2011	10	3	0.016	0.001	0.543	0.663*	4.909	2.65	65	0.621	0.704	10.000	3.98
Walby	MatSu	61.371	149.212	2011, 2012	10	7	0.123	0.009	0.653	0.718*	5.727	2.92	58	0.617	0.751	9.875	4.41
Wolf	MatSu	61.166	149.166	2010, 2011	10	4	0.215	0.015	0.696	0.673	5.273	2.66	50	0.646	0.711	9.625	4.13
Cornelius	MatSu	61.374	149.152	2012	10	3	0.263	0.02	0.701	0.682	5.545	2.83	13	0.586	0.718	5.875	4.12
Loberg	MatSu	61.333	149.141	2012	10	2	0.013	0.001	0.687	0.7	5.909	2.77	81	0.602	0.716	10.500	4
Falk	MatSu	61.332	149.031	2010	10	1	0	0	0.781	0.674*	5	2.66	86	0.632	0.745	11.000	4.23

N hosts: number of three-spined stickleback hosts; H_d : haplotype diversity in lake; N_d : nucleotide diversity among haplotypes; H_o : observed heterozygosity; H_e : expected heterozygosity; N_a : average number of alleles across 11 loci; A_r : average rarefied allelic richness across loci.

*Indicates significant deviations from Hardy–Weinberg equilibrium.

MICROSATELLITE GENOTYPING AND ANALYSIS FOR STICKLEBACK HOSTS

All fish were genotyped at the following 11 microsatellite (μ sat) loci: stn33, stn46, stn67, stn171, stn168, stn174, stn120, stn12, stn79, 7033PBBE, and stn38 (Largiadèr *et al.*, 1999; Peichel *et al.*, 2001). We used a QIAGEN Multiplex PCR kit (Qiagen, Inc., Valencia, CA, USA) to amplify loci in mixes of three or four loci using PET, HEX, TAMRA, or FAM labelled forward primers. Reaction conditions for μ sat amplifications were 95 °C for 15 min, followed by 35 cycles of 94 °C for 30 s, 57–61 °C (depending on the primer mix) for 90 s, 72 °C for 60 s, and a final step of 60 °C for 30 min. Genotype fragment analysis was conducted on an ABI 3730xl DNA Analyzer (Applied Biosystems, Foster City, CA, USA). Resulting data were edited and scored using GeneMarker v. 2.6.3 (Softgenics, State College, PA, USA). Micro-checker v. 2.2.3 (Van Oosterhout *et al.*, 2004) was used to screen the μ sat data for null alleles and scoring errors.

We used Genalex v. 6.501 to calculate measures of genetic diversity including number of alleles, observed and expected heterozygosity, as well as departures from Hardy–Weinberg equilibrium. Rarefied allelic richness was calculated in SPAGeDi v. 1.5 (Hardy & Vekemans, 2002). Genetic differentiation was assessed between pairs of lake populations by estimating F_{ST} , as well as across hierarchical levels (i.e. among regions, among lakes, among individuals), and among sampling years, by means of AMOVA in Arlequin v. 3.5. The significance of pairwise F_{ST} values was determined from 10 000 bootstrap replicates and sequential Bonferroni correction (Holm, 1979). Genetic structure was also assessed using a Bayesian analysis of population differentiation in STRUCTURE v. 2.3.4 (Pritchard, Stephens & Donnelly, 2000). We implemented the admixture model with correlated allele frequencies (Falush, Stephens & Pritchard, 2003) and no a priori information on the geographic origin of individuals. Ten independent analyses were run for each value of K , from $K = 1$ to $K = 15$, consisting of 200 000 Markov chains with a prior burn-in of 200 000 chains. The uppermost level of genetic structure was selected based on the rate of change of the log probability of data [$\ln P(X|D)$] between successive K values (ΔK) following Evanno, Regnaut & Goudet (2005) as implemented in CLUMPAK (Kopelman *et al.*, 2015).

MICROSATELLITE DATA ANALYSIS FOR CESTODE PARASITES

Data from eight μ sat loci (Samonte-Padilla & Kalbe, 2012) were derived from Sprehn *et al.* (2015) for 805 *S. solidus* parasites isolated from fish hosts that were sequenced and genotyped for the current study.

Genetic structure was assessed by calculating pairwise population F_{ST} values with significance determined from 10 000 bootstrap replicates and sequential Bonferroni correction (Holm, 1979). We also conducted AMOVAs to test for hierarchical differentiation among lakes and regions, as well as among sample years. All analyses were performed in Arlequin v. 3.5. Additionally, the number of genetic clusters that minimizes Hardy–Weinberg and linkage disequilibrium was inferred using the Bayesian clustering algorithm of STRUCTURE v. 2.3.4 following Sprehn *et al.* (2015).

HOST–PARASITE COMPARISONS

We employed three approaches to test for associations between genetic variation in stickleback hosts and genetic variation in cestode parasites. We first conducted AMOVAs to test for hierarchical differentiation according to fish host mtDNA lineage, mtDNA haplotype, and nuclear genotypic cluster in Arlequin v. 3.5. Using GENODIVE v. 2.0b27 (Meirmans & Van Tienderen, 2004). We then performed Mantel tests (Mantel, 1967) between geographic distances between lakes and pairwise F_{ST} and Φ_{ST} matrices for parasites and hosts, respectively. We also created a binary temporal matrix (using zero to indicate pairs of lakes sampled in the same year and 1 to indicate that lakes were sampled in different years) and performed Mantel tests between the temporal matrix and pairwise F_{ST} and Φ_{ST} matrices for parasites and hosts, respectively. We also performed partial Mantel tests between pairwise F_{ST} or Φ_{ST} matrices of parasites and hosts (following Martinez *et al.*, 1999), correcting for a third matrix of either geographic distances or sampling year. Finally, we performed canonical redundancy analysis (RDA), which is a multiple linear regression method that, unlike Mantel tests, enables disentanglement of the relative contribution of host genetic variation, spatial variation, and temporal variation in driving genetic structure of parasites. We used the matrix of STRUCTURE q values of cluster assignment proportion for each parasite individual as dependent variables. The explanatory variables were a spatial matrix S (x and y coordinates, and polynomial transformations thereof x^2 , y^2 , xy , x^2y , xy^2 , x^3 , y^3), a host genetic matrix H , which was divided into two categories: mtDNA variation and μ sat variation, and sampling year. The mtDNA variation matrix included either *cytb* mtDNA lineage (ENAC or TNPC) or haplotype data and the μ sat matrix included individual q values from STRUCTURE. Thus, two separate analyses were performed, one considering only stickleback lineage as the mtDNA explanatory variable, and one considering *cytb* haplotype as the mtDNA explanatory variable. For each set of explanatory variables we

performed global RDAs and report adjusted regression coefficients of multiple determination (R^2_{adj}). We used an automatic forward selection procedure to improve the fit of each model and to reduce the likelihood of type I errors. In addition to this, we calculated variance partitioning (Borcard, Legendre & Drapeau, 1992) to estimate the variance (in genetic structure of the parasite) uniquely explained by the host ($H|S \cup T$), by space ($S|H \cup T$), or by time ($T|H \cup S$), and assessed the significance of each variable. Variance decomposition was performed only if either the global spatial, the host, or the time models were significant. In the partial RDA models, covariables were included only if significant in any of the models. RDA analyses were performed in R v. 3.2.0 (R Core Team, 2015) using the packages *vegan* (Oksanen *et al.*, 2015) and *SoDA* (Chambers, 2013).

RESULTS

STICKLEBACK HOST GENETIC VARIATION

In total, 20 haplotypes were recovered from 161 three-spined sticklebacks based on sequence variation across 499 bp of the *cytb* gene. Haplotypes were

representative of the ENAC and TNPC lineages that were divergent by 14 substitutions; we recovered 14 ENAC and six TNPC haplotypes (Supporting Information, Figs S1 and S2). Both lineages occurred in all three regions, but differed in relative frequencies and only co-occurred in six lakes (Fig. 2 and Supporting Information, Fig. S2). The TNPC lineage was more common in BB, whereas the ENAC lineage was more common in the Kenai and the MatSu regions (Fig. 1 and Supporting Information, Fig. S2). Walby Lake and Falk Lake, both from the MatSu region, contained the highest and lowest number of haplotypes, respectively ($h = 7$ and $h = 1$) (Table 1). Also, Walby Lake contained three exclusive haplotypes (H11, H16, and H19), whereas all other lakes contained no more than one exclusive haplotype. Haplotype diversity indices ranged from $H_d = 0.296$ in Aleknagik Lake to $H_d = 0.00$ in Falk Lake (Table 1). Pairwise Φ_{ST} values ranged from -0.008 between Engineer Lake and Walby Lake to 0.94 between Lower Ohmer Lake and Falk Lake, with the majority (58 of 105; 55.2%) of the comparisons significant after sequential Bonferroni correction ($\alpha' = 0.001$) (Supporting Information, Table S1). Genetic differentiation among regions explained a higher percentage

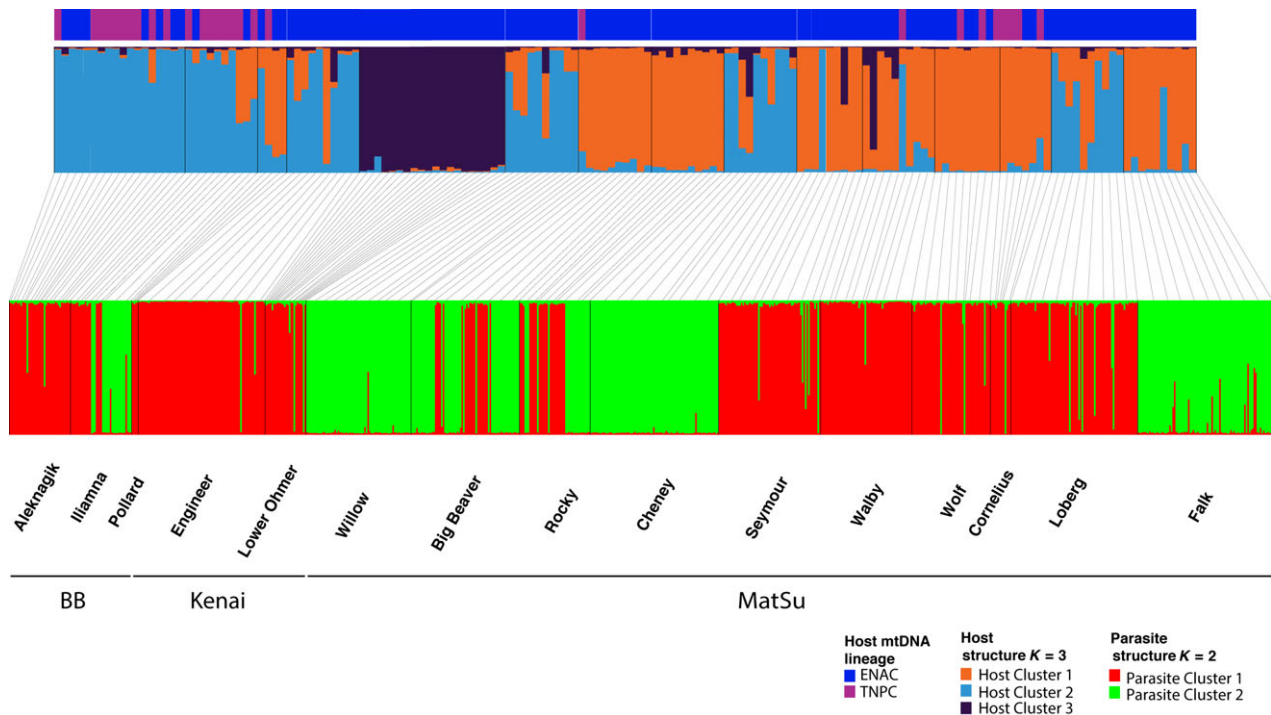


Figure 2. Mitochondrial lineage assignment for all three-spined stickleback individuals, and Bayesian analysis of microsatellite variation of three-spined sticklebacks and *S. solidus*. In the top graph, host mitochondrial lineage is indicated in pink (TNPC) and blue (EPAC). In the middle graph, STRUCTURE assignment proportions for host genotypic clusters at $K = 3$ are indicated in orange (Cluster 1), light blue (Cluster 2), and purple (Cluster 3). In the bottom graph, assignment proportions for *S. solidus* clusters at $K = 2$ are indicated in red (Cluster 1), and green (Cluster 2). Thin grey lines connect the host graph to the parasite graph indicating how many parasites were sampled from each individual host.

of variation ($\Phi_{CT} = 0.377$, $P < 0.01$) than among lakes within regions ($\Phi_{SC} = 0.249$, $P < 0.001$) (Table 2). Host mitochondrial variation over time (i.e., sampling year) also was significant ($\Phi_{ST} = 0.059$, $P = 0.045$).

Microsatellite-based estimates of genetic variation did not always parallel variation observed at the *cyt b* gene. For example, individuals from Cheney Lake, in the MatSu region, exhibited the highest number of alleles ($N_a = 7.27$) and allelic richness ($Ar = 2.98$) (Table 1). Conversely, the lowest value of rarefied allelic richness ($Ar = 2.24$) and the lowest observed heterozygosity ($H_o = 0.471$) were found in Lower Ohmer Lake in the Kenai Peninsula (Table 1).

Most pairwise comparisons among lakes (95 out of 105; 90.5%) were significant after sequential Bonferroni correction ($\alpha' = 0.004$), with F_{ST} values ranging from 0.049 between Rocky Lake and Big Beaver Lake in MatSu, to 0.309 between Lower Ohmer Lake and Pollard Lake in Kenai (Supporting Information,

Table S1). When fish were grouped by lakes and regions, a higher percentage of variation was explained by lakes within regions ($F_{SC} = 0.132$, $P < 0.001$) than among regions ($F_{CT} = 0.044$, $P < 0.01$) (Table 2). Host microsatellite variation over time also was significant ($F_{ST} = 0.036$, $P < 0.001$).

Bayesian clustering analysis revealed pronounced genetic structuring among lakes and among regions (Fig. 2). We also found marked disagreement between mitochondrial and nuclear genetic structure, with no clear correspondence between lineages and clusters (Fig. 2). The ΔK values calculated from the posterior probabilities revealed two comparable peaks at $K = 3$ and $K = 5$, indicating the presence of hierarchical genetic substructure (Fig. 2 and Supporting Information, Fig. 3). Subsequent analyses relied on the more conservative and highest level of genetic structure of $K = 3$, at which most lakes were assigned with a high proportion of individuals to a single cluster. Cluster 1 was the most abundant

Table 2. Summary of AMOVA results for three-spined stickleback hosts and their cestode parasites

	μsat				mtDNA			
	% of variation	F	95% CI	P	% of variation	F	95% CI	P
<i>Gasterosteus aculeatus</i>								
Among years	3.6	$F_{ST} = 0.036$	0.023, 0.049	$< 0.001^*$	5.85	$\Phi_{ST} = 0.059$	−0.003, 0.122	0.045*
Among host clusters	9.56	$F_{ST} = 0.096$	0.076, 0.116					
Among regions (BB, Kenai, MatSu)	4.36	$F_{CT} = 0.044$	0.021, 0.066	$< 0.01^*$	37.7	$\Phi_{CT} = 0.377$	0.329, 0.418	$< 0.01^*$
Among lakes within regions	12.59	$F_{SC} = 0.132$	0.101, 0.154	$< 0.001^*$	15.48	$\Phi_{SC} = 0.249$	0.200, 0.301	$< 0.001^*$
Among all lakes	83.05	$F_{ST} = 0.170$	0.138, 0.188	$< 0.001^*$	46.81	$\Phi_{ST} = 0.532$	0.504, 0.566	$< 0.001^*$
<i>Schistocephalus solidus</i>								
	μsat							
	% of variation	F	95% CI	P				
Among years	1.82	$F_{ST} = 0.018$	0.001, 0.048	$< 0.001^*$				
Among parasite clusters	9.2	$F_{ST} = 0.092$	0.002, 0.251					
Among regions (BB, Kenai, MatSu)	1.49	$F_{CT} = 0.015$	0.001, 0.031	0.131				
Among lakes within regions	4.94	$F_{SC} = 0.050$	0.008, 0.128	$< 0.001^*$				
Among all lakes	93.58	$F_{ST} = 0.064$	0.017, 0.130	$< 0.001^*$				
Among host haplotypes (H1–H20)	1.01	$F_{CT} = 0.010$	0.045, 0.019	$< 0.001^*$				
Among host fish within haplotypes	6.76	$F_{SC} = 0.068$	0.024, 0.142	$< 0.001^*$				
Among all host fish	92.22	$F_{ST} = 0.078$	0.030, 0.159	$< 0.001^*$				
Among host lineages (ENAC, TNPC)	0.51	$F_{CT} = 0.005$	0.000, 0.012	0.033*				
Among host fish within lineages	7.42	$F_{SC} = 0.075$	0.028, 0.153	$< 0.001^*$				
Among all host fish	92.08	$F_{ST} = 0.079$	0.030, 0.163	$< 0.001^*$				
Among host clusters (Cluster 1, Cluster 2, Cluster 3)	0.28	$F_{CT} = 0.003$	0.001, 0.004	0.065				
Among host fish within clusters	7.37	$F_{SC} = 0.074$	0.027, 0.151	$< 0.001^*$				
Among all host fish	92.35	$F_{ST} = 0.077$	0.028, 0.156	$< 0.001^*$				

* denote significant values ($P < 0.05$).

cluster found predominantly in the MatSu region, with the exception of Pollard Lake in Kenai and a few individuals elsewhere (Fig. 2). Cluster 3, on the other hand, was the rarest cluster and only found in Lower Ohmer Lake in Kenai and in one individual in Walby Lake in MatSu. Individuals from Engineer Lake in Kenai were predominantly assigned to Cluster 2, which was most prevalent in BB (Fig. 2). At $K = 5$, greater admixture among clusters was observed, especially in the lakes from the MatSu region (Supporting Information, Fig. 3). Lower Ohmer Lake and Aleknagik Lake, however, clearly represented two distinct clusters from all other lakes. Most individuals in Engineer Lake in Kenai also were assigned to the same cluster as Cheney Lake and Loberg Lake in MatSu (Supporting Information, Fig. 3).

CESTODE PARASITE GENETIC VARIATION

A significant proportion of variation was explained among samples in lakes within regions ($F_{SC} = 0.050$, $P < 0.001$), whereas the proportion of genetic variation attributable to differences among regions was not significant (Table 2). Parasite microsatellite variation was significant among years ($F_{ST} = 0.018$, $P < 0.001$). Bayesian clustering analysis performed on parasites isolated from fish used in this study recovered the same pronounced genetic structure as described in Sprehn *et al.* (2015) (Fig. 2). Support was found for two clusters, with individuals in most lakes assigned to one genetic cluster, with the exceptions of Iliamna, Big Beaver, and Rocky. Individuals also were assigned with high proportion to a single genetic cluster (i.e., 97% of individuals were assigned to their cluster with $q > 0.8$) (Fig. 2). The two

clusters were nearly equal in frequency (0.57 and 0.43 for Cluster 1 and 2 respectively).

HOST-PARASITE COMPARISONS

A significant proportion of parasite nuclear genetic variance was explained by host mitochondrial lineage ($F_{CT} = 0.005$, $P = 0.033$), though more variation was attributed to host fish within lineages ($F_{SC} = 0.075$, $P < 0.001$) (Table 2). When we considered host mitochondrial haplotypes (Table 2), a significant percentage of variation was explained among host haplotypes ($F_{CT} = 0.010$, $P < 0.01$), though more variation was attributed to individual host fish within haplotypes ($F_{SC} = 0.068$, $P < 0.001$). Similarly, when we considered nuclear host genotypic clusters (Table 2), microsatellite variation of parasites among host fish within clusters was greater than variation among host clusters ($F_{SC} = 0.074$, $P < 0.001$ and $F_{CT} = 0.003$, $P = 0.065$).

Mitochondrial genetic differentiation of hosts and nuclear genetic differentiation of parasites were significantly and positively correlated with geographical distances between populations (Table 3). A stronger correlation was observed for parasites (Mantel's $r = 0.421$, $P = 0.006$) than for fish (Mantel's $r = 0.216$, $P = 0.019$) (Table 3). Nuclear genetic differentiation of hosts and geographic distance recovered a non-significant trend of isolation by distance (Mantel's $r = 0.4$, $P = 0.076$) (Table 3). Neither parasite nuclear genetic differentiation nor host mitochondrial genetic differentiation was significantly and positively correlated with the temporal matrix (Table 3). A significant correlation between host Φ_{ST} and parasite F_{ST} was recovered (Mantel's $r = 0.159$, $P = 0.047$), whereas a non-significant trend was

Table 3. Mantel and partial Mantel test results

Matrix 1	Matrix 2	Matrix 3	Mantel's r	R^2	P
Geographic	Host Φ_{ST}		0.216	0.047	0.019*
Geographic	Host F_{ST}		0.4	0.16	0.076
Geographic	Parasite F_{ST}		0.421	0.177	0.006*
Host Φ_{ST}	Parasite F_{ST}		0.159	0.025	0.047*
Host F_{ST}	Parasite F_{ST}		0.221	0.049	0.071
Host Φ_{ST}	Parasite F_{ST}	Geographic	0.077		0.62
Host F_{ST}	Parasite F_{ST}	Geographic	0.063		0.611
Host Φ_{ST}	Temporal		-0.07	0.005	0.242
Parasite F_{ST}	Temporal		-0.199	0.039	0.033*
Host Φ_{ST}	Parasite F_{ST}	Temporal	0.149		0.161

Euclidean geographic distance vs. mitochondrial and microsatellite-based measures of pairwise genetic differentiation; Host genetic differentiation vs. parasite genetic differentiation; Partial Mantel tests of host vs. parasite differentiation controlling for a third matrix of Euclidean distance between lakes; Host and parasite genetic differentiation vs. sampling years; Host genetic differentiation vs. parasite differentiation controlling for a third temporal matrix of sampling years. * denote significant values ($P < 0.05$).

recovered between host F_{ST} and parasite F_{ST} (Mantel's $r = 0.221$, $P = 0.071$) (Table 3). Neither correlation was significant when controlling for geographic distance or time (Table 3).

The global RDA showed that a combination of space, host genetic variation (microsatellite and *cytb* haplotypes), and time explains 73.638% ($R^2_{adj} = 0.725$) of the variance in genetic structure of the parasite (Supporting Information, Table S2). All nine geographical variables were significant, as well as 15 host genetic variables (14 mtDNA and 1 μ sat) and one temporal variable, yielding a reduced model that explained 71.434% ($R^2_{adj} = 0.705$) of the total variance ($P < 0.001$). Spatial variables alone explained 65.927% of the total variance ($R^2_{adj} = 0.655$), with a unique contribution ($S|H \cup T$) of 35.110% ($R^2_{adj} = 0.354$). The combination of mtDNA haplotypes and μ sat host genetic variation explained 18.582% ($R^2_{adj} = 0.171$) of the variance in parasite genetic structure, of which the unique (and significant) contribution of mtDNA haplotype ($H|S \cup T$) was 3.800% ($R^2_{adj} = 0.034$). The unique contribution of host μ sat variation, conversely, was not significant (Supporting Information, Table S2). Similar results, although with lower explained variances, were obtained when we only included lineage identity as mtDNA variation. In this case, the reduced model for all variables explained 67.979% ($R^2_{adj} = 0.675$) of the variance, and the unique contribution of mtDNA variation was smaller (0.300%, $R^2_{adj} = 0.003$) but still significant (Supporting Information, Table S2). The unique contribution of time was significant ($P < 0.001$) and explained 1.260% ($R^2_{adj} = 0.006$) of parasite variance when considering host haplotype identity and 1.890% ($R^2_{adj} = 0.014$) when considering host lineage identity (Supporting Information, Table S2).

DISCUSSION

We evaluated the relative contributions of host genetic variation, time, and geographic factors in shaping genetic variation of the parasite *Schistocephalus solidus* infecting three-spined sticklebacks. Our results indicate that geography, time, and genetic variation in stickleback hosts were significant predictors of parasite genetic variation. Geographical factors appear to be the strongest drivers of population genetic structure in *S. solidus*. Indeed, significant genetic structure among lakes in *S. solidus* parasites and stickleback hosts indicates a strong influence of geography in shaping the distribution of genetic variation in both species. Discordant population genetic structure between hosts and parasites, however, suggests that different factors

govern host and parasite population differentiation (McCoy *et al.*, 2005; Keeney *et al.*, 2009). This might reflect differences in movement potential, but it is also possible that the observed infection patterns instead reflect differences in vulnerability to infection or admixture among host lineages (Lescak *et al.*, 2015; Kalbe *et al.*, 2016; Lagrue *et al.*, 2016). Evidence of temporal variation and associations with host haplotypes also raise the possibility that distributional differences in the hosts which *S. solidus* preferentially infect underlie the observed patterns of population structure.

THE INFLUENCE OF GEOGRAPHY AND HOST DISPERSAL ON PARASITE POPULATION STRUCTURE

Biogeographic patterns of genetic variation were not consistent between *S. solidus* and three-spined stickleback hosts across the study region. The distribution of haplotype and genotypic variation followed a longitudinal pattern in three-spined sticklebacks, where the TNPC lineage predominated in western lakes and the ENAC lineage was more frequent in eastern lakes, particularly east of Engineer Lake. This pattern is consistent with evidence of broader biogeographic asymmetries across the northern Pacific Rim (Lescak *et al.*, 2015). Though *S. solidus* parasites were found to be genetically differentiated into two clusters, there does not appear to be an analogous longitudinal distributional pattern within the study region (Sprehn *et al.*, 2015). Both clusters were distributed nearly equally in the eastern and western areas of the study region. Further comparisons across the northern Pacific Rim will be necessary to characterize biogeographical variation in *S. solidus*, and to confirm whether patterns parallel those found in three-spined stickleback hosts.

The discordance in patterns observed in three-spined sticklebacks and *S. solidus* corresponds to expectations based on differences in rates of evolution, life history, dispersal potential, and barriers to gene flow (McCoy *et al.*, 2005; Keeney *et al.*, 2009; Sprehn *et al.*, 2015). Three-spined sticklebacks are known to exhibit rapid adaptive genomic differentiation (Gibson, 2005; Marques *et al.*, 2016). Although some evidence suggests that *S. solidus* keep pace with stickleback hosts (Kalbe *et al.*, 2016), it is possible that differences between sticklebacks and *S. solidus* could give rise to discordant patterns of genetic variation (Dybdahl & Lively, 1996; Gandon *et al.*, 2008; McCoy *et al.*, 2005; Keeney *et al.*, 2009). Discordance also could reflect less genetic drift in *S. solidus* populations with larger effective population sizes and higher genetic diversity compared to three-spined sticklebacks (Keeney *et al.*, 2009). The observed patterns could, however, be a result of

differences in movement potential among intermediate stickleback and definitive avian hosts (Smyth, 1962; Keeney *et al.*, 2009; Sprehn *et al.*, 2015). This follows the premise that, due to the dependency of parasites on hosts for movement (Nadler, 1995; Criscione, Poulin & Blouin, 2005), gene flow in parasites is expected to correspond to the movement potential of the most vagile host (Jarne & Théron, 2001; Prugnolle *et al.*, 2005). Nevertheless, there appears to be more genetic structure in Alaskan *S. solidus* populations than expected from the vagility of definitive avian hosts (Sprehn *et al.*, 2015).

Genetic differentiation in *S. solidus* may be explained by gene flow constraints due to geographic or behavioral barriers limiting the potentially extensive dispersal by avian hosts, local specificity to three-spined stickleback host lineages or ecotypes, or a combination of both factors (Criscione *et al.*, 2005; Sprehn *et al.*, 2015; Kalbe *et al.*, 2016). Differential infection patterns observed between parasite clusters and host haplotypes and lineages may be partially a result of constrained dispersal if birds tend to distribute parasites within the same lake or between lakes in close proximity (Sprehn *et al.*, 2015). Alternatively, if birds distribute parasites evenly (or randomly) among lakes, differences in specificity of the two parasite clusters may be driving the observed pattern of differential infection. Genetic drift or selective pressures from the environment (external to the host) could also lead to differentiation of isolated parasite populations, irrespective of host specificity (Sprehn *et al.*, 2015; Lagrue *et al.*, 2016).

HOST-MEDIATED INFECTION

Parasites that interact with a succession of hosts are unlikely to exhibit concordant genetic variation with any particular host because of diffuse coevolutionary interactions (Lajeunesse & Forbes, 2002; Lagrue *et al.*, 2016). Nonetheless, our results indicate that stickleback host genetic identity plays a small but significant role in shaping population genetic variation in *S. solidus*. This finding is consistent with experimental evidence of reciprocal adaptive specificity between stickleback host and *S. solidus* parasite populations (Kalbe *et al.*, 2016). Results of the RDA analyses indicate that host μ sat variation does not contribute to the structure of genetic variation in *S. solidus*, whereas a small but significant portion of *S. solidus* nuclear genetic variation corresponds to stickleback mtDNA lineages and haplotypes independently of geography and time (Supporting Information, Table S2). Frequencies of Cluster 1 and Cluster 2 parasites differed significantly between ENAC and TNPC host lineages as well as between rare (i.e., haplotypes found in $\leq 1\%$ of fish sampled) and

common host haplotypes. Sticklebacks belonging to the TNPC lineage, which is less prevalent in Alaska than the ENAC lineage, were infected by a higher frequency of Cluster 1 parasites than Cluster 2 parasites whereas ENAC sticklebacks were infected by nearly equal proportions of Cluster 1 and Cluster 2 parasites ($\chi^2 = 9$, d.f. = 1, $P < 0.01$) (Supporting Information, Table S3 and Supporting Information, Fig. 2B). Also, rare haplotypes were disproportionately infected by Cluster 1 parasites whereas common haplotypes were infected nearly equally by both clusters ($\chi^2 = 12$, d.f. = 1, $P < 0.001$) (Table S3 and Supporting Information, Fig. 2B).

The relatively small effect of host lineage on parasite genetic variation may not reflect the full extent of host specialization if adaptive specificity is trait specific (Kalbe *et al.*, 2016; Lagrue *et al.*, 2016). A study of the trematode parasite *Coitocaecum parvum* and its crustacean host *Paracalliope fluviatilis* revealed, for example, that local adaptation of individual traits may occur without pronounced codivergence (Lagrue *et al.*, 2016). It is also possible that the full extent of host specialization is not evident because of genomic admixture following secondary contact between stickleback evolutionary lineages. The biogeographic progression from ENAC to TNPC lineages transitions across south-central Alaska (Lescak *et al.*, 2015). Consistent with evidence of genomic admixture across stickleback contact zones elsewhere (e.g., Lucek *et al.*, 2010), mito-nuclear discordance in our study area is suggestive of genomic admixture among stickleback lineages. Admixture could disrupt the genomic configuration (and thus integrity) of individual-level attributes that determine host specialization. If so, then stronger signatures of host-driven genetic concordance might be evident in comparisons among populations with little to no mito-nuclear discordance, which are most likely located farther southeast and southwest of our study area (Lescak *et al.*, 2015). Additionally, admixture may have given rise to evolutionary novelty (e.g., Lucek *et al.*, 2010) that promotes specialization to admixed hosts independent of parental evolutionary lineages. Experimental infection trials have found, however, that admixed stickleback hosts exhibit an intermediate infection rates and growth of *S. solidus* from either parental area of origin (Kalbe *et al.*, 2016).

Although parasites might adapt in response to genetic, immunological, or ecological differences between populations of stickleback hosts, it is also possible that hosts differ in susceptibility to infection (Haglund *et al.*, 1992; Ortí *et al.*, 1994; Kitano, Mori & Peichel, 2007; Kitano *et al.*, 2011; Eizaguirre *et al.*, 2011; Kalbe *et al.*, 2016; Lagrue *et al.*, 2016). Reciprocal infection trials comparing sympatric and

allopatric populations of stickleback and *S. solidus* indicate that infection probably reflects local adaptation to the sympatric counterpart in the host and the parasite, corresponding to parasite specific resistance in the host and optimal virulence in the parasite (Kalbe *et al.*, 2016). Differences in susceptibility among populations of three-spined stickleback may possibly be due to variation in immunological responses to infection by *S. solidus* (Franke *et al.*, 2014; Kalbe *et al.*, 2016; Lagrue *et al.*, 2016). For example, differences in both parasite communities and MHC allele pools have been found between river and lake ecotypes of three-spined sticklebacks (Eizaguirre *et al.*, 2011). Non-immunological responses also differ among stickleback populations. For example, a study of three-spined stickleback in the United Kingdom found differences in reproductive behaviours and sexual development in infected stickleback males from different populations (MacNab, Katsiadaki & Barber, 2009). Alaskan three-spined stickleback females also show variable pathologies when infected with *S. solidus*: certain populations show fecundity reduction resulting from nutrient theft, whereas other populations show fecundity compensation (Heins, 2012; Heins & Baker, 2014). These or other differences in host response may have arisen as a consequence of genetic drift (Eizaguirre *et al.*, 2011) or local adaptation (Kalbe *et al.*, 2016), but they might also map to evolutionary lineage, reflecting adaptation of ENAC and TNPC stickleback hosts to infection by *S. solidus* in general or in response to different genotypic clusters (Kalbe *et al.*, 2016). Pronounced phenotypic and behavioural differences have been documented between ENAC and TNPC sticklebacks and between other differentiated stickleback groups (Deagle *et al.*, 1996; Kitano *et al.*, 2007, 2011; Ravinet *et al.*, 2014) that could give rise to differences in susceptibility.

POTENTIAL FOR FREQUENCY-DEPENDENT SELECTION

The Red Queen Hypothesis is often referred to when describing host–parasite coevolutionary dynamics for adaptive traits that afford each species an advantage over the other (Van Valen, 1973). A modified hypothesis proposes that host–parasite coevolution might occur via cyclical, time-lagged, negative frequency-dependent selection (Hamilton, 1980; Bell & Smith, 1987; Hamilton, Axelrod & Tanese, 1990; Lively, 1996). Under this hypothesis, rare host phenotypes would hold an advantage over common host phenotypes because parasites would adapt to infect the most common phenotype (Lively, 1996). Host–parasite frequency-dependent selection has been described in natural and laboratory populations of the snail *Potamopyrgus antipodarum* infected by a trematode

parasite, where parasites track common clones with a time-lagged pattern, resulting in rare host clones being less infected than common clones (Dybdaahl & Lively, 1998). Comparisons of novel combinations simulated in a laboratory setting to naturally occurring host clone–parasite combinations of *Daphnia magna* and its bacterial parasite *Pasteuria ramosa* also suggest that parasites track common host clones in nature (Carius, Little & Ebert, 2013). Similarly, we found that one genetic group (Cluster 2) of *S. solidus* parasites tends to infect the most common host haplotypes, whereas the other group (Cluster 1) more often infects low frequency haplotypes. Cluster 2 parasites might be better adapted to the most common host haplotypes and thus have a narrower host range than Cluster 1 parasites. If parasite clusters do preferentially infect host haplotypes based on local prevalence (Kalbe *et al.*, 2016), then distributional differences and temporal variation in host haplotypes could underlie the observed pattern of parasite population structure. This hypothesis could be tested through experimental infection trials (Kalbe *et al.*, 2016; Lagrue *et al.*, 2016). A study of long-term infection patterns – arguably longer than 3 years, given that coevolutionary interactions can vary over different spatial and temporal scales (Thompson & Cunningham, 2002; McCoy *et al.*, 2005) – by the two parasite clusters might also prove informative; under frequency-dependent selection, Cluster 2 parasites would be expected to track the most common host haplotypes in a time-lagged manner.

CONCLUSION

Evidence of geographic and host-mediated population structure suggests that *S. solidus* parasites respond to conditions external and internal to their host (Gandon *et al.*, 1998; Thompson & Cunningham, 2002; Orsini *et al.*, 2013). This suggestion is consistent with prior work showing that geography and host identity contribute to the structure of genetic variation in other widely distributed cestode parasites like *Ligula intestinalis* (Štefka *et al.*, 2009). Although our findings indicate that the conditions imposed by intermediate stickleback hosts are less important than those related to geography and dispersal (Mazé-Guilmo *et al.*, 2016), it is possible that further work might uncover stronger signatures of adaptive specificity. Determining whether ENAC and TNPC sticklebacks (or particular haplotypes therein) present different internal environments with which parasites must contend would be particularly informative (Gandon *et al.*, 1998; Kalbe *et al.*, 2016). Experimental infection trials could be an effective

approach for determining host susceptibility and host specialization in parasites with complex life cycles (e.g., Kalbe *et al.*, 2016; Lagrue *et al.*, 2016). Comparative analysis of functional loci governing host susceptibility and parasite virulence might also shed further light on outcomes of genomic admixture and experimental evidence of reciprocal adaptive specificity between three-spined sticklebacks and *S. solidus* parasites (Eizaguirre *et al.*, 2011; Franke *et al.*, 2014; Kalbe *et al.*, 2016).

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article:

Figure S1. Neighbour-joining tree of unique haplotypes recovered from three-spined sticklebacks.

Figure S2. (A) Three-spined stickleback haplotype network coded by region. (B) Three-spined stickleback haplotype network coded by parasite genetic cluster.

Figure S3. Three-spined stickleback STRUCTURE diagnosis at $K = 5$.

Table S1. Pairwise Φ_{ST} and F_{ST} by Lake.

Table S2. Redundancy analysis (RDA).

Table S3. Frequencies of Cluster 1 and Cluster 2 parasites infecting rare vs. common host haplotypes and ENAC vs. TNPC host lineages.

SHARED DATA

Data available from the Dryad Digital Repository: doi:10.5061/dryad.b1c06 (Strobel *et al.*, 2016).